

MURSHID VP

Backend Engineer · Python · FastAPI · Microservices · AI/ML
+91 9995332723 · murshidvp.dev@gmail.com · Kerala, India

PROFESSIONAL SUMMARY

Backend Engineer with 5+ years building production-grade Python and FastAPI services for global fintech clients including Citibank, HSBC, and US healthcare customers. Deep expertise in REST API design, microservices architecture, async programming with Celery/Redis, and PostgreSQL query optimization. Proven track record shipping secure, scalable systems — including AI-powered document pipelines processing millions of files at 97–98% accuracy and an RFQ platform cutting procurement turnaround by ~95%. Experienced with Docker, CI/CD, DevOps practices, and Agile delivery.

EXPERIENCE

Senior Backend Engineer / AI Engineer · KG Information Systems Ltd, 2021 – Present
Coimbatore

Backend Architecture & API Development

- Architected and shipped production FastAPI and Flask microservices to serve AI and ML models at scale — handling RESTful API design, versioning, and lifecycle management.
- Built and managed secure REST APIs and async task pipelines using Celery distributed workers and Redis caching — powering I/O-heavy inference, batch workloads, and real-time procurement workflows.
- Implemented JWT-based authentication, authorization middleware, and secure coding practices aligned with fintech compliance standards across Citibank and HSBC deployments.
- Designed and optimized PostgreSQL schemas, indexes, and CTEs on multi-million-row tables — reducing feature-extraction queries from minutes to sub-second response times.
- Engineered multiprocessing and multithreaded Python pipelines cutting batch document-AI processing from hours to minutes across millions of files.
- Containerized all backend services via Docker and deployed through CI/CD pipelines; maintained Linux-based production environments.

Fintech & Enterprise Document Intelligence

- Deployed OCR + ML extraction pipelines for Citibank and HSBC processing 100K+ documents/month at ≥97% accuracy with automated digital-signature validation — reducing manual review by >90%.
- Built AI-powered IRS K-1/K-3 tax form extraction system (250+ fields) for DOZ, USA — cutting per-return processing from days to minutes at ~97% accuracy, unblocking peak-season throughput.

AI & ML Integration

- Integrated OpenAI (GPT), Google Gemini, and Llama into production Python backends — chatbot interfaces, document Q&A, and intelligent extraction with prompt engineering and graceful model fallback.
- Designed AI-powered supplier discovery for EVOSTEL: top 5 suppliers per part via cascading geographic priority (Saudi Arabia → GCC → Global) — cutting procurement turnaround by ~95%.
- Built end-to-end ML pipelines for healthcare/insurance regression and classification models — feature engineering, hyperparameter tuning, and deployment driving measurable accuracy gains.
- Designed NLP sentiment-analysis service (TF-IDF + Word2Vec) surfacing actionable insights from unstructured customer-feedback streams.

Data & Observability

- Built Streamlit/Matplotlib/Seaborn dashboards delivering real-time procurement and operational insights to enterprise stakeholders.

- Performed EDA, data cleaning, and feature analysis using Pandas and NumPy across banking, healthcare, and procurement datasets.

SELECTED PROJECTS

EVOSTEL · AI-Powered Procurement Platform

End-to-end RFQ workflow with multi-stage approvals and real-time analytics. FastAPI backend, Celery workers, Redis caching, Dockerized deployment. AI supplier discovery returns top 5 suppliers per part with cascading geographic priority — cutting procurement turnaround by ~95%.

Enterprise Document AI · Citibank & HSBC

OCR + ML extraction pipelines processing millions of banking documents at 97–98% accuracy with automated signature validation. Reduced manual review by >90% across two global banking clients.

K-1/K-3 Tax Form Extraction · DOZ, USA

AI extraction system mapping 250+ IRS schedule fields. FastAPI + Celery backend; cut per-return processing from days to minutes at ~97% accuracy.

LLM Chatbot & NLP Workflows

Multi-model LLM integration (OpenAI GPT, Gemini, Llama) into production Python services — chatbot interfaces, document Q&A, intelligent extraction with prompt engineering and graceful model fallback. Companion TF-IDF + Word2Vec sentiment pipeline for customer-feedback classification.

TECHNICAL SKILLS

- **Backend & APIs:** Python, FastAPI, Flask, Django, REST APIs, WebSockets, Microservices, Async Programming
- **Databases:** PostgreSQL (query optimization, indexing, CTEs), Redis, MySQL, ORM tools
- **Distributed Systems:** Celery, Redis, Multiprocessing & Multithreading, Event-driven Architecture
- **DevOps & Infra:** Docker, CI/CD Pipelines, Linux, Git, Cloud Deployment, Agile/Scrum
- **Security:** JWT, OAuth, Secure Coding Practices, Fintech Compliance Standards
- **AI / LLMs:** OpenAI GPT, Google Gemini, Llama, Prompt Engineering, Multi-Model Orchestration
- **ML & NLP:** Regression, Classification, Feature Engineering, TF-IDF, Word2Vec, Sentiment Analysis
- **Data & Analytics:** Pandas, NumPy, EDA, Matplotlib, Seaborn, Streamlit
- **Frontend:** React (intermediate), HTML, CSS, JavaScript, jQuery, AJAX

ACHIEVEMENTS

- **Best Performer Award, Marvel.ai (2025)** — recognized for client-onsite delivery impact.
- **Spotlight Award, Marvel.ai (2024).**
- Consistently delivered AI, ML, and document-intelligence systems at ~98% accuracy and >90% manual-effort reduction across banking, healthcare, insurance, tax, and procurement domains.

EDUCATION

BE Computer Science · Hindustan College of Engineering & Technology, Coimbatore